ETALE QUASI-COMPACT SEPARATED U OVER A HENSELIAN PAIR SPLITS INTO FINITE PART AND PART AWAY FROM Z

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 05K0: sources/stacks-chow/tags.txt, source lines 7184–7184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09XI: sources/stacks-chow/tags.txt, source lines 12531–12531. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let (A, I) be a henselian pair. Let $U \to \mathrm{Spec}(A)$ be a quasi-compact, separated, étale morphism such that $U \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I) \to \mathrm{Spec}(A/I)$ is finite. Then*

$$U = U_{fin} \amalg U_{away}$$

where $U_{fin} \to \mathrm{Spec}(A)$ is finite and U_{away} has no points lying over Z.

Proof. By Zariski's main theorem, the scheme U is quasi-affine. In fact, we can find an open immersion $U \to T$ with T affine and $T \to \mathrm{Spec}(A)$ finite, see More on Morphisms, Lemma 05K0. Write $Z = \mathrm{Spec}(A/I)$ and denote $U_Z \to T_Z$ the base change. Since $U_Z \to Z$ is finite, we see that $U_Z \to T_Z$ is closed as well as open. Hence by More on Algebra, Lemma 09XI we obtain a unique decomposition $T = T' \amalg T''$ with $T'_Z = U_Z$. Set $U_{fin} = U \cap T'$ and $U_{away} = U \cap T''$. Since $T'_Z \subset U_Z$ we see that all closed points of T' are in U hence $T' \subset U$, hence $U_{fin} = T'$, hence $U_{fin} \to \mathrm{Spec}(A)$ is finite. We omit the proof of uniqueness of the decomposition. □

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